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Deccan flood basalts at the Cretaceous/Tertiary boundary?

Vincent Courtillot [1], Jean Besse [1], Didier Vandamme [1], Raymond Montigny [2], Jean-Jacques Jaeger [3] and Henri Cappetta [4]

[1] *Laboratoire de Paléomagnétisme et Géodynamique, Institut de Physique du Globe, Place Jussieu, 75005 Paris (France)*
[2] *Laboratoire de Géochimie, Institut de Physique du Globe, Rue Descartes, Strasbourg (France)*
[3] *Laboratoire de Paléontologie des Vertébrés, Université P et M Curie, Place Jussieu, 75252 Paris (France)*
[4] *Institut des Sciences de l'Evolution, Université des Sciences et Techniques du Languedoc, Place Bataillon, 34060 Montpellier Cedex (France)*

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Joint consideration of new paleomagnetic, paleontological and geochronological data from the Deccan continental flood basalts in India and critical discussion of earlier results lead us to suggest that volcanic activity may have lasted less than 1 Ma, thus possibly ranking as one of the largest volcanic catastrophes in the last 200 Ma Available data are best satisfied if volcanism spanned the Cretaceous/Tertiary boundary, followed shortly afterwards by rifting of the Arabian Sea These results point out the need for further work which may help in choosing between "external" and "internal" models of the Cretaceous/Tertiary boundary events

1. Introduction

The Deccan traps in India form one of the largest known occurrences of continental flood basalts (Fig. 1) When one takes into account the flows that may lie off the coast to the west and those that are likely to have been removed by erosion, it is found that the trap basalts may have covered an area in excess of 10^6 km^2 [1] The thickness of the lava pile grows from the northern and eastern boundaries to a value possibly in excess of 2000 m inland of the Bombay Coast, along the western Ghats The total volume of lava is thus likely to have exceeded 10^6 km^3 [1]. Yet, despite decades of extensive geological work, still relatively little is known about the internal structure, age, total duration and ultimately origin of the Deccan volcanism

As far as internal structure is concerned, the lava pile consists of rather thick (10–160 m) flat lying flows, with dips seldom exceeding 1°, except near Bombay where dips of 5° towards the sea are observed Therefore, altitude has often been used as a simple first-order stratigraphic indication This ignores the possibility of faults, which are not easy to identify because of the nature of basalts and the conditions of outcrops Also, one must be aware that non-measurable dips not exceeding 1° may lead to altitude differences of many hundreds of meters over the distances of several hundred kilometers which are typically encountered in the Deccan. Attempts have been made (e.g. [2]), and are still being made (K. Cox, personal communication, 1985, [3,4]), to decipher the lava stratigraphy with petrological and geochemical markers, but so far the overall structure remains very poorly known, except in the southwestern Ghats It is generally accepted [1], though by no means definitely established, that the oldest flows are exposed to the east, where the series is thinnest, and that outcropping flows become gradually younger towards the west, until one reaches maximum thickness (and altitude) in the western Ghats. From there on, one reaches the coast through a series of steps, with seaward bending and possibly normal faults roughly parallel to the coast itself.

The absolute age and total duration of the volcanism have been the subject of a number of K-Ar studies [5–9] and one ^{39}Ar-^{40}Ar study [10]. Some authors [1,2,5] concluded from those studies that volcanic activity lasted from ~ 100 to ~ 40 Ma ago This includes, however, the small outcrops of the Rajmahal traps to the east of the Deccan, which have ages around 100 Ma and are

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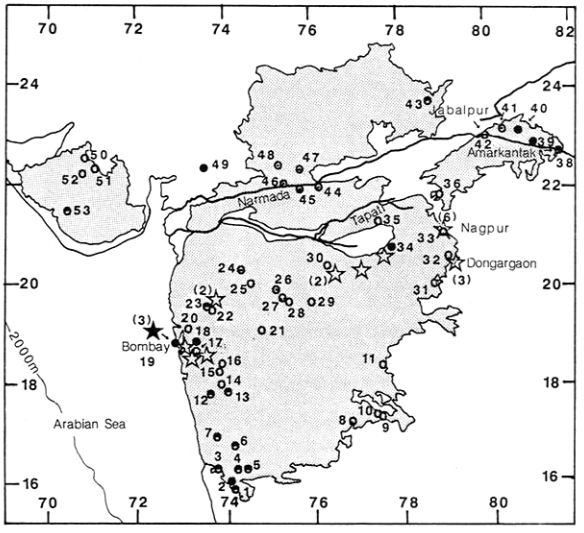


Fig 1 Present outcrops of the Deccan basalts (shading) with our paleomagnetic sites indicated as stars (with the number of nearby sites given in parentheses) and the available magnetostratigraphic sections reviewed by Rao et al [2] as circles Full symbols indicate normal magnetic polarities, open symbols reversed polarities, half-filled symbols are for normal reversed (N_1/R_1) magnetostratigraphies and dotted circles are for remaining sections discussed specifically in the text

clearly distinct from them. Still, the absolute ages from the Deccan traps s.s range from over 85 Ma [10] to less than 30 Ma [9]

Further hopes of establishing a better age bracket come from paleomagnetic and magnetostratigraphic studies. As early as 1968, McElhinny [11] compared what was known at the time of the probable sequence of magnetic polarities in the Deccan traps with early versions of the réversal time scale and concluded that much of the volcanic activity might have taken place in only 5 Ma. In 1984, we initiated a new sampling program in the Deccan, in the hope of obtaining further constraints on its origin. The first paleomagnetic, geochronologic and paleontological results along a traverse from Nagpur to Bombay (Fig 1) are reported in this paper, and earlier available data are reviewed in the light of the new data

2. Paleomagnetic results

Twenty-two sites were sampled on the traverse from Nagpur to Bombay; most sites consist of 6–8 cores sampled in a single flow over a few tens to hundreds of meters. The twenty-two sites correspond to ten distinct geographic locations From the above considerations, it is possible that the oldest (i.e lowermost) flows are located in the east of the traverse. Over 120 specimens cut from the cores were subjected to alternating field and stepwise thermal demagnetizations in field-free space. Measurements were performed on a CTF cryogenic and Molspin spinner magnetometer Standard procedures (Zijderveld vector plots, Kirshvink least-squares determination of magnetization direction, Hall's technique of remagnetization circles) were used to recover characteristic primary thermo-remanent magnetizations A single component magnetization with coercitive fields between 20 and 80 mT and unblocking temperatures between 200 and 570°C was isolated in 15 out of 22 sites Demagnetization temperature and a f magnetic fields indicate that some form of (titano-) magnetite is the principal magnetic carrier, with restricted occurrences of hematite and goethite. A two-component magnetization was found in 7 sites, but the high-temperature component could easily be recovered, in 3 sites using the technique of remagnetization circles A detailed report of the paleomagnetic study will be found elsewhere [12,13] and the final results are summarized in Table 1 For the 21 sites (site 13, which is quite distant from the mean, was excluded), the mean virtual geomagnetic pole is located at 290.5°E and 32.5°N, with a confidence interval of 5° at the 95% probability level The resulting paleolatitude reduced at Nagpur is 29°S (±4°) at the time of lava emplacement

The most important result of this study is that 18 out of the 21 sites, i e 85% of the total, have a reversed magnetization (Fig. 1 and Table 1). This completes and confirms previous findings that the remanent magnetization of the Deccan traps is predominantly reversed. In early compilations of paleomagnetic results from the traps, Wensink and Klootwijk [14] and Wensink [15] listed results, 80% of which corresponded to reversed magnetization

Because of the lack of good stratigraphic control in the Deccan, one might fear that this magnetic polarity bias might to some extent result from sampling a small number of flows, this could arise from unfortunate conjugate effects of erosion, altitude and undetected faults. This concern

TABLE 1

Paleomagnetic results from the Nagpur-Bombay traverse

Locality		No	N	D (°)	I (°)	K or Q (*)	α_{95} (°)
Dongargaon	20°31′N, 79°17′E	01	9	172 0	47 4	107 7	5 0
		02A	4	158 1	43 6	2 2 *	–
		02B	5	151 1	28 8	8 1 *	–
		03	6	155 3	53 9	34 7	11 5
Nagpur	21°10′N, 79°12′E	04–05	8	150 6	45 7	6 8 *	–
		06	4	155 7	43 8	390 5	4 3
		07	3	149 1	54 2	105 5	12 1
		08	5	138 8	40 5	203 2	5 4
		09	3	136 8	37 1	2458 4	2 5
		11	4	131 2	44 0	310 3	5 2
Amravati	20°58′N, 77°50′E	13	6	178 5	30 9	146 5	5 6
Akola	20°40′N, 77°05′E	14	4	164 0	48 0	209 1	6 4
Buldana	20°31′N, 76°18′E	15	4	129 9	42 0	84 6 *	–
		16	6	154 9	47 2	788 7	2 4
Ighatpury	19°41′N, 73°38′E	18	5	140 2	30 4	119 3	7 0
		20	6	161 4	44 7	562 2	2 8
Bombay	18°53′N, 72°51′E	21	7	331 8	−45 4	337 9	3 3
		22	3	328 6	−37 1	174 8	9 4
		23	5	319 8	−57 6	21 1	17 0
"Road" [a]	18°53′N, 73°20′E	25	5	168 4	56 3	78 2	8 7
Lonawala	18°46′N, 73°44′E	26	9	135 6	60 4	188 0	3 8
Khandala	18°44′N, 73°29′E	27	5	131 8	47 2	35 9	12 9
Deccan		all but 13	21 [b]	147 5	46 8	50 9	4 6

Paleolatitude of Nagpur 28 8°S ± 4°

N = number of samples,
D = declination of the magnetization direction,
I = inclination of the magnetization direction,
K = Fisher's precision parameter,
Q = Hall's precision parameter ($\times 10^{-6}$),
α_{95} = semi-angle of the cone of 95% confidence
[a] "Road" site on the Bombay-Poona road (78 km from Bombay)
[b] Number of sites

can partly be relieved by a consideration of published magnetostratigraphic sections in the traps. Many such sections are now available [2,15,16], although it is not always easy to have access to and assess the degree of reliability of the original data We have projected 40 sections on a roughly E-W line, more precisely running across the traps perpendicular to the Bombay coast, which is likely to correspond to the dominant direction of normal faulting (Fig. 2) This direction is roughly that of opening of the Arabian Sea, and has long been inferred to be parallel to the main fissures through which the Deccan basalts were emitted [1] In Fig. 2 we have simply used altitude for the ordinate. The longest sections shown in Fig. 2 are up to almost 1000 m thick (i.e. about half of the maximum thickness of the lava pile or more) and they span 1650 m overall. Section 13 contains over 40 flows, sections 46 and 47 over 20 flows each Out of a combined 11,720 m of sections, 8770 m (or 75%) are reversed. More important, very few sections show more than one reversal, and none shows more than 2 Most of the sections that display one reversal only begin with a reversed part at the bottom, with normal flows on top. Seven out of 40 sections deserve special mention. Two sections (24 and 26) display a normal flow, less than 30 m thick, within a long, entirely reversed section. The flow in section 26, coming from a study of Bhimasankaram and Pal [17], is listed as intermediate or unstable by Verma and Mital [18] The average declinations, at 290 and 309° respectively for the only two available sites, are quite anomalous [19]. The flow in section 24

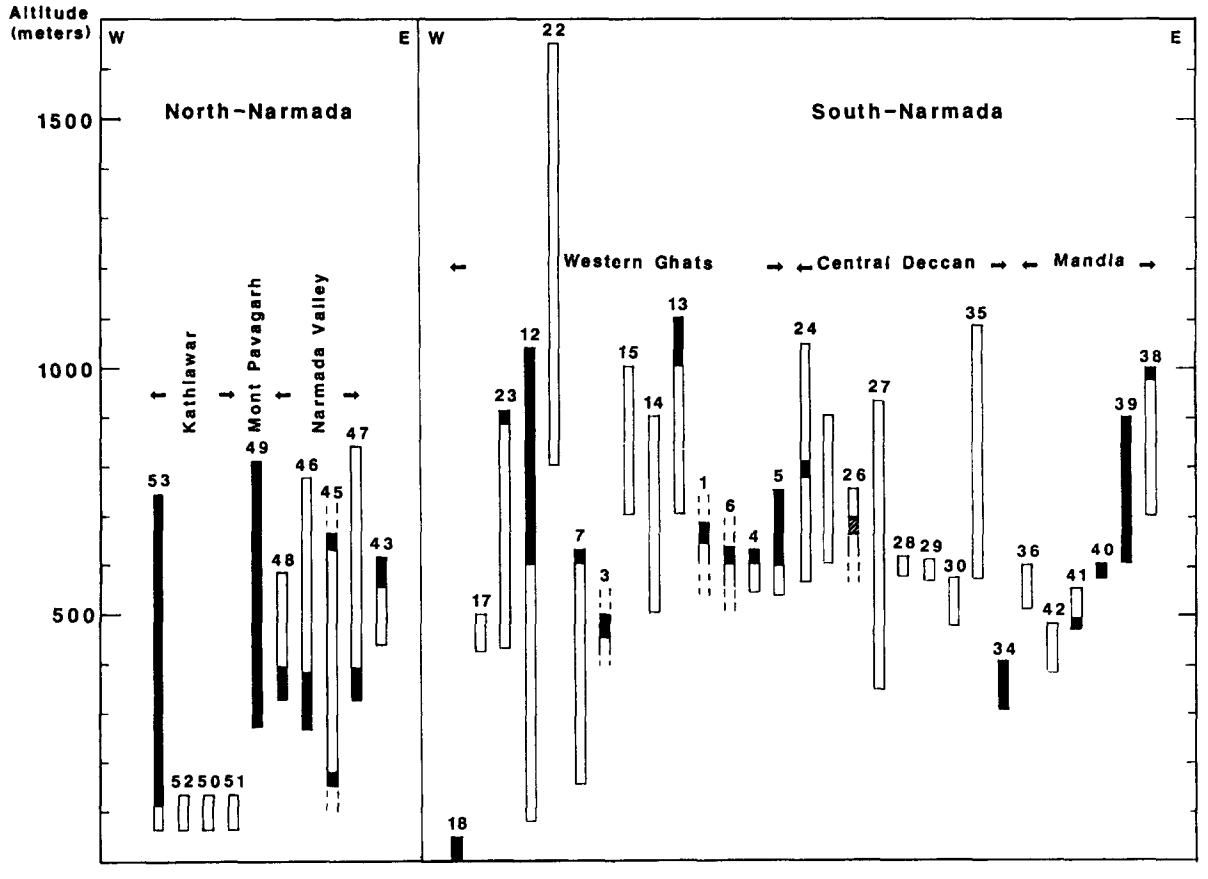


Fig 2 Magnetostratigraphic sections (magnetic polarity as a function of altitude) projected on a profile with azimuth N80°E, perpendicular to the Bombay coast Normal polarity in black, reversed in white, locations in Fig 1 (mostly after data found in Rao et al 's [2] review, with some corrections from original publications)

[19] has both an anomalous low inclination of 21° and a very large 95% confidence interval of 28°. Nearby sections (25 and 27) are entirely reversed, with no indication of a normal interval at similar altitudes (Figs 1, 2) One section (41) on the road from Jabalpur to Amarkantak (Fig 1) begins with a 15 m normal flow (Kundam section [20]) Finally, four sections in the Narmada Valley (44, 46, 47, 48), and particularly two on its northern bank [2,21] display 6 normal flows (for a total thickness of 120 m) below a thick reversed pile

The typical Deccan stratigraphy may thus consist of a thin (often absent) normal sequence, a thick intermediate reversed sequence and an upper normal sequence. When comparing the altitude at which the reversed/upper normal boundary occurs, one must be aware of the effects of very modest dips and undulations over very large distances For instance, sections 38 and 39, which would imply a vertical offset of at least 400 m, are separated by 75 km and are compatible with our proposed reversal sequence if a westward dip of order 0 3° exists east of Jabalpur In the same way, a dip of 0 6° can account for the 800 m offset between sections 34 and 35, and a dip of 0.3° is sufficient between sections 12 and 22 In the latter case, the dip is of the same sense (southwards) and amount found from recent independent geochemical studies by Devey and Lightfoot [4] and Beane et al [3]. The 800 m (or more) offset between sections 22 and 23, which are separated by 20 km, requires a somewhat larger average dip of order 2°, or alternately a fault. Although unfrequent and uneasy to observe, normal faults are indeed thought to occur in the area of the western Ghats

Overall, available paleomagnetic results (Table

1, Figs 1 and 2) support the hypothesis that Deccan trap volcanism may have covered no more than three polarity intervals (or chrons) This discussion strongly strengthens a similar hypothesis made in 1979 by Klootwijk [16]. A large fraction of the activity (on the order of 80%) appears to have taken place during the middle reversed interval

3. Paleontological observations

It has been known for over a century that the youngest sediments immediately underlying the traps (the so-called infratrappean sediments), for instance the Lameta beds in the vicinity of Jabalpur, contain dinosaur remains. These show considerable resemblance to the Maestrichtian South American dinosours and have therefore been considered as most probably of Santonian to Maestrichtian (i.e. Late Cretaceous) age [22] More recently, in association with the new paleomagnetic and geochronological sampling, the tooth of a ray (Fig 3) was recovered from the same beds. This tooth can be assigned to the Myliobatidae, the morphological features and dimensions of this tooth, though it is incomplete, are identical to those of the teeth described as *Igdabatis* from the Maestrichtian of Niger [23] Jain and Sahni [24] have recovered remains of *Igdabatis* and other fishes from the Lameta beds near Pisdura, the age of which is Upper Cretaceous to Lower Paleocene according to these authors Since *Igdabatis* is a characteristic element of the Maestrichtian fauna of Niger, a Maestrichtian age can be proposed for the Lameta beds near Jabalpur and by extension to other Indian deposits where this genus occurs Above the first trap flows, intertrappean sediments are found in thin layers. These intertrappean beds have long been considered as Paleocene on the basis of their flora More recently, an analysis of Charophytes from the Nagpur intertrappean [25] has revealed an age younger than Campanian but older than Dano-Montian (Paleocene) (see also [26]). The same beds have yielded carnivorous theropod (dinosaur) teeth [27,28] and dinosaur egg shell remains [28] These observations have received further confirmation from a study of planktonic foraminifera found about 30 km to the southwest of Rajahmundry [29] both the sediments below and above the very

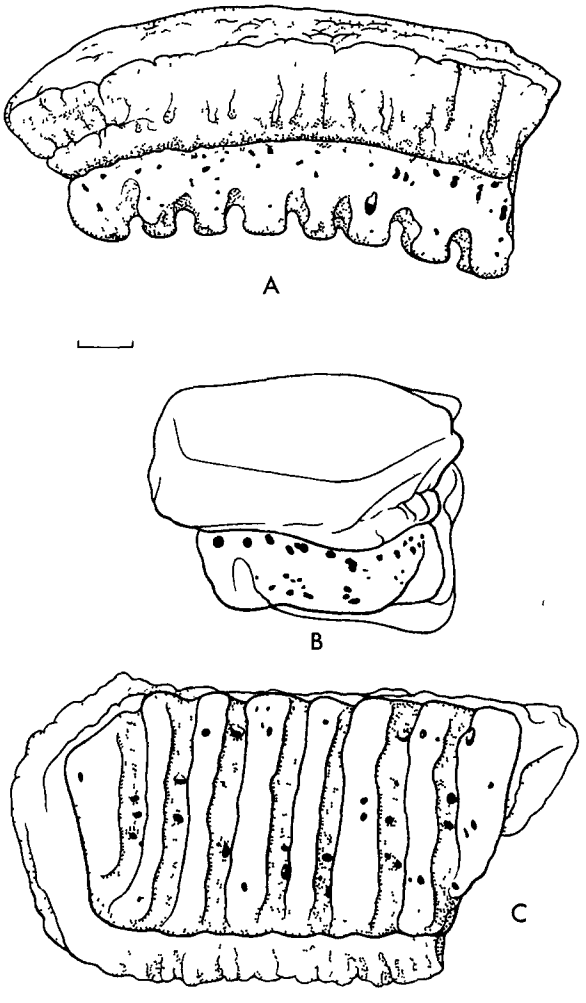


Fig 3 Incomplete tooth of *Igdabatis* cf *sigmodon* (JAB 1) (A) Lingual view, (B) lateral view, (C) basal view The scale bar represents 1 mm

first trap level have yielded Maestrichtian marine microfaunas. Just above the traps, which are very thin near Rajahmundry, marine microfaunas are attributed to zone P2 of the lower Paleocene [30].

Our own paleomagnetic results from the lowermost flow in Dongargaon, south of Nagpur, which is interlayered in the Maestrichtian Lameta sediments, show only reversed polarity. This situation is therefore identical to that at Nagpur. This suggests that in some areas Deccan volcanism must have begun within a reversed interval of the Maestrichtian, but before the beginning of the Paleocene. In other areas (Jabalpur, Narmada), there is some evidence for one to a few normal

flows in the earliest stages. Volcanism probably ended prior to Paleocene paleontological zone P2.

4. Absolute age determinations

Fourteen samples coming from six of the sites sampled for paleomagnetism were dated using the K-Ar method. The samples were selected in the eastern half of the traverse, where the oldest flows were expected from structural and stratigraphic considerations. The experimental technique was that described in Westphal et al. [31] Microscopic examination of thin sections revealed some evidence for hydrothermal alteration, and magnetic and non-magnetic fractions were isolated and measured separately The results are listed in Table 2 and displayed in Fig. 4. The more altered samples are found to correspond to the non-magnetic fractions, with younger apparent ages and larger uncertainties (open dots in Fig. 4). Ages for the less altered samples range from 55 to 66 Ma, with 1σ error bars between 2 and 4 Ma.

Previous determinations have been listed in Fig. 4. Open symbols again correspond to more altered samples or less reliable age determinations, according to the original papers. The two oldest ages are ^{39}Ar-^{40}Ar plateau ages from Kaneoka [10]. They both come from samples (listed as altered or less reliable) collected in the vicinity of Bombay There appears to be some doubt as to their reliability and even stratigraphic position: the samples might even not belong to the Deccan s s (M Kono, personal communication, 1986) Restricting ourselves only to the more reliable determinations (33 of them), we find no age older than 67 Ma There is a peak in the number of data between 60 and 67 Ma, confirming the early proposal of Wellman and McElhinny [6]. The histogram of age determinations decreases almost monotically to 40 Ma. Almost all authors interpret ages younger than 55 Ma as due to argon loss resulting from some amount of alteration (e.g. [7]) Indeed, plagioclase is the major K-bearing mineral which is used in the age determinations This mineral is quite sensitive to minor reheating and fluid circulation, which may account for the scatter in ages. Many ages are younger than the maximum limit, which can be considered as probably

TABLE 2

K-Ar ages

Locality	No	HF	Part	Mineral	K_2O (wt %)	100 rad ^{40}Ar / Total ^{40}Ar	Rad ^{40}Ar	Age (Ma)
Dongargaon	01	tr	m	Pl	0 310	66 0	3 008	66 2± 3 9
	02		m	Pl	0 394	53 0	3 412	59 2± 2 8
	02	tr *	nm	Pl	0 168	14 6	1 044	42 6± 7 5
Nagpur	0 4	*	nm	Pl	0 118	6 5	0 6812	39 7±12 0
	04	tr *	nm	Pl	0 128	12 0	0 7885	42 3± 8 0
	04	tr	nm	Pl+Ox	0 160	45 0	1 441	61 5± 5 0
	08	tr *	nm	Pl	0 138	6 9	0 9684	48 2±10 0
	08	tr	m	Pl	0 168	38 0	1 518	61 7± 5 0
Buldana	15	tr	nm	Pl	0 150	34 0	1 300	59 2± 6 5
	15	tr *	m	Pl	0 229	53 0	2 028	60 5± 3 5
	16	*	m	Pl+Ox	0 337	43 0	2 539	51 6± 2 5
	16	tr	nm	Pl	0 371	20 0	2 972	54 8± 3 5
	16	tr	m	Pl	0 528	74 0	4 928	63 7± 2 5
	16	tr	m	Pl+Ox	0 409	72 0	3 965	66 1± 3 0

$\lambda_\epsilon = 0\,581\times10^{-10}$ a^{-1}, $\lambda_\beta = 4\,962\times10^{-10}$ a^{-1}, ^{40}K/total K $= 1\,167\times10^{-4}$

HF Treatment with hydrofluoridric acid * Occurrence of calcite, eliminated by HF treatment and/or significant amounts of altered plagioclases, tr = treated samples

Part Studied part of samples (after separation) m = magnetic, nm non-magnetic

Mineral Studied mineral Pl = plagioclase, Ox = oxide

Uncertainty in each age mean at the 1σ (68%) confidence level

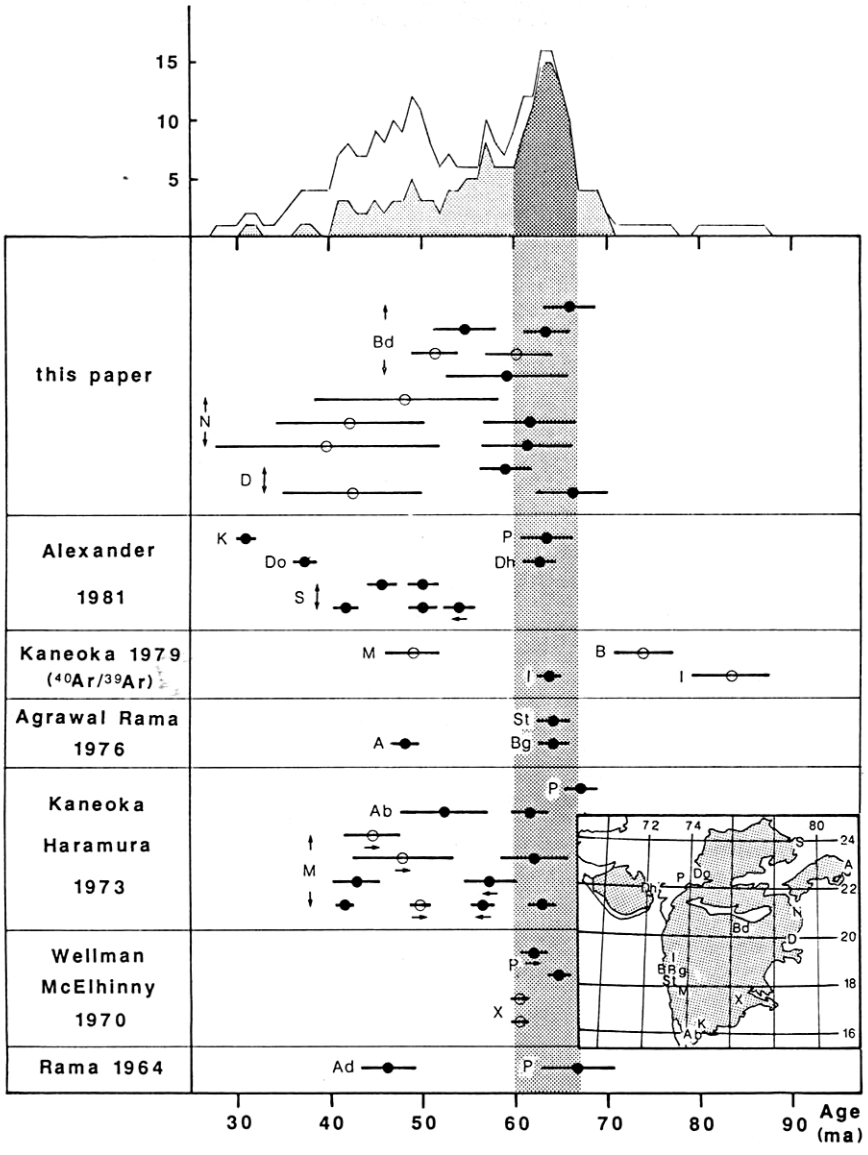


Fig 4 K-Ar ages for the Deccan traps, with approximate location (inset), from this study (see also Table 2), and also Alexander [9], Kaneoka ([10], ^{39}Ar-^{40}Ar), Agrawal and Rama [8], Kaneoka and Haramura [7], Wellman and McElhinny [6] and Rama [5] Open symbols are for altered samples according to the original papers, 1σ error bars are indicated On the top of the figure are two histograms for the age distributions (shaded for non altered samples only, unshaded for all samples) Arrows in the Kaneoka and Haramura [7] study indicate K/Ar ages which are in severe contradiction with stratigraphical position in a given lava pile (arrows toward right, stratigraphically older, arrows toward left, stratigraphically younger)

very close to the true age We note that there is some amount of correlation between younger and less accurate ages and stronger magnetic overprints (e.g sites 2, 4 and 15), although there is also some evidence of alteration in sites which yielded very good single component magnetic directions (e.g. sites 1, 8 and 16).

5. Timing and duration of volcanism

The three types of data described above, i e. paleomagnetic, paleontological and geochronologic, provide independent information and can be combined to yield some constraints on the age of the Deccan volcanism. Paleontological observa-

tions would indicate that basalt emission began within the (upper) Maestrichtian, i.e. no later than 65–67 Ma ago, depending on which geological time scale is used Unless there is an (as yet undetected) systematic bias in K-Ar towards younger ages, the data shown in Fig. 4 would imply that volcanism began no earlier than 67 Ma ago Put together, both approaches imply that the first eruptions must have taken place rather precisely at 66 ± 2 Ma, with a conservative error estimate

Paleomagnetic data, on the other hand, would imply that the total duration of Deccan volcanism did not exceed (part of) three successive magnetic chrons. In Fig. 5 we compare six recent reversal time scales. All of them indicate that the reversal rate was rather high around 66 Ma ago, with no systematic polarity bias Thus, the 80% polarity bias found in Deccan in itself suggests that the total duration of volcanism cannot have exceeded 3 Ma. Moreover, we know that the Cretaceous/Tertiary (KTB) boundary occurs within chron 29R [32], hence the combined paleomagnetic/paleontological data in the Dongargaon/Nagpur area tell us that the reversed chron corresponding to most of the Deccan trap activity cannot be younger than 29R Chron 30R lasted only ~ 90,000 years and appears to be far too short to be acceptable Chron 31R is too old to satisfy the geochronological data, if one follows the reversal time scales of Ness et al [33], Lowrie and Alvarez [34] or Berggren et al [35] It might be barely acceptable if the LaBrecque et al [36] or Cox [37] time scales were used It would be acceptable if

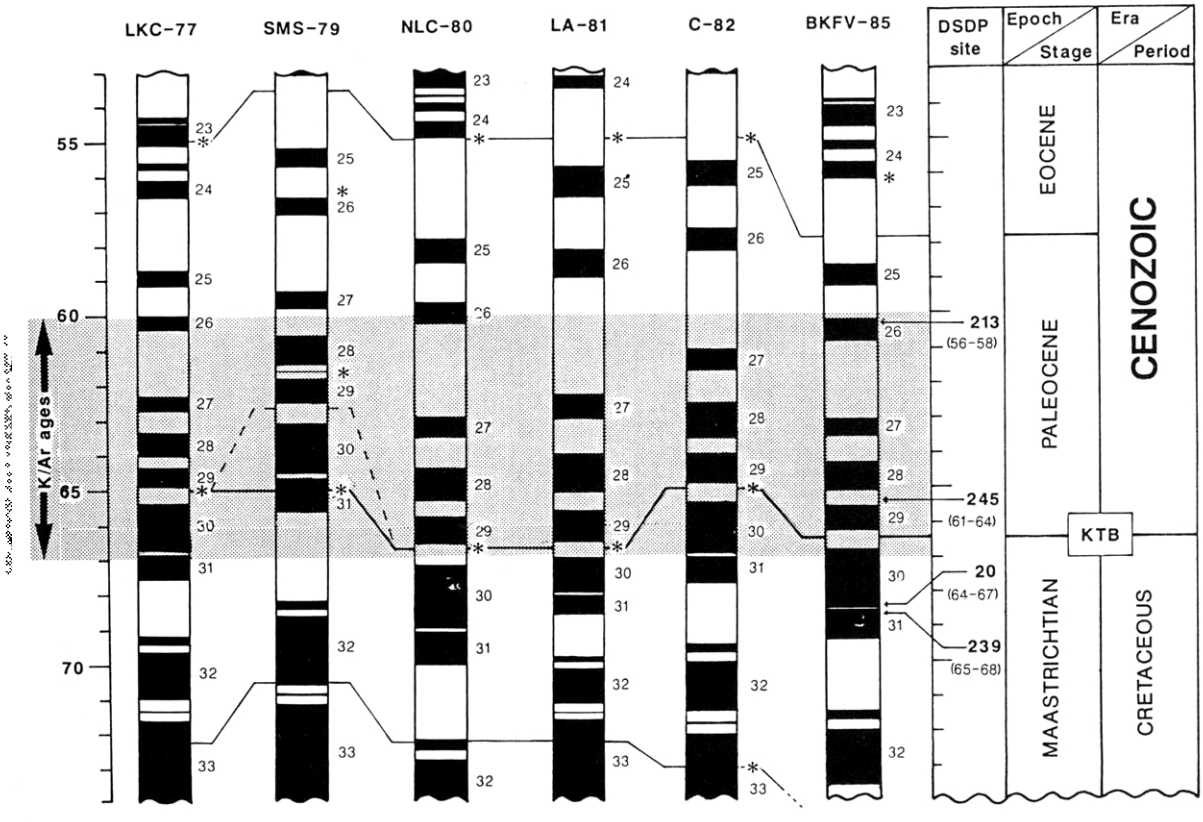


Fig 5 A comparison of six recent magnetic reversal time scales (LKC 77 [36], SMS 79 [38], NLC 80 [33], LA 81 [34], C 82 [37], BKFV 85 [35] Normal (reversed) polarities are shown in black (white) Chron 29R is at the Cretaceous/Tertiary boundary The range of K-Ar ages of Deccan basalts from this study is shaded Biostratigraphic tie-points used in the various time scales are shown as stars (most of them revisions of the ages given by Hardenbol and Berggren [30]), and DSDP sites are listed with their biostratigraphic age ranges

the Schlich et al [38] scale is valid This last scale relies heavily on the determination of the bottom age for DSDP hole 239 [38]. Moreover, it does not include use of the new K/Ar decay constants [39] and involves a 2 Ma discrepancy between chron 29R and the biostratigraphical KTB [30].

Should the correlation between the reversed Deccan interval and chron 31R be accepted (and we have seen that this was only marginally possible), the total duration of Deccan volcanism could be as long as 3 Ma (only 2 Ma if the LA-81 scale is used), if the entire sequence of a reversed chron separated by two normal chrons is valid (31'N/31R/31N). This would imply that all of the K-Ar ages are systematically more than 2–4 Ma in error However, the previous discussion leads us to propose that the most likely correlation, which satisfies available data, is with chron 29R Because of the length of normal chron 30N, the upper bound for the duration of volcanism remains essentially unchanged (Fig. 5). However, if we accept that the ratio of the total thickness of reversed/normal flows is roughly proportional to the time spent in reversed/normal geomagnetic polarities (i.e. 80%), then the upper bound becomes less than 1 Ma (actually about 0 5 Ma), the exact value depending on which time scale is used (the sequence being. end of 30N, 29R, beginning of 29N) We therefore see that the total duration of Deccan trap basaltic volcanism might not have exceeded 1 Ma and that this effusive event might have spanned anomaly 29R, which contains the KTB

Our correlation is in complete contradiction with the recent conclusions of Rao et al. [2], who tentatively identified the lowermost Deccan normal interval found at the base of the Narmada section over a thickness of 6 flows with the Cretaceous Long Normal superchron (which lasted from 110 to 80 Ma ago). Paleontological data on the Lameta beds are by themselves sufficient to discard this correlation The total duration of Deccan volcanism may be significantly shorter than that estimated by McElhinny [11], and even by Kono [40], who argued from a combined statistical analysis of the reversal time scale and of lava flow emplacement for an upper bound of 3 Ma for a large part of the total section. On the other hand, Officer and Drake [41] arrived at a low figure (they propose 0 43–0 88 Ma). Although, Officer and Drake's line of reasoning is close to ours from a magnetostratigraphic point of view, their detailed evaluation is not given and they used only a limited amount of rather old data. In particular, they did not have the observations required to propose a reasonably safe correlation of Deccan magnetostratigraphy with the reversal time scale

6. Deccan flood basalts and the Cretaceous/Tertiary boundary

The combined analysis of our new data and a review of previous work thus leads us to suggest that the Deccan trap eruption may have been a truly catastrophic event, probably one of the largest such events in the last 200 Ma of Earth history Average extrusion rates could have been in excess of 1 $km^3\ a^{-1}$ for a duration of about 1 Ma For comparison, the global surface spreading rate is 3 5 $km^2\ a^{-1}$ [42,43], i e. 2–4 $km^3\ a^{-1}$ in volume of basaltic layer formation, also for comparison, the average output of lavas from the Hawaiian hot spot is about 0 1 $km^3\ a^{-1}$ (e.g. [44]).

Moreover, we suggest that the eruption may have coincided, with a ~ 1 Ma resolution, with the KTB. This would add to the series of apparently anomalous events that took place at that time, particularly the mass extinctions and the Ir concentrations discovered by Alvarez et al. [45] We wish to note here that another geodynamic event occurred shortly after Deccan volcanism and the KTB, namely the opening of the Arabian Sea and rifting of the Seychelles-Mascarene (partly continental) oceanic plateau. Indeed, seafloor spreading between Madagascar-Africa and the Seychelles-Mascarene-India continent stopped at the time of anomaly 28 [46] and jumped north, initiating formation of the Arabian Sea just prior to anomaly 27 ([47], see Fig. 6) This jump would thus have occurred roughly 2 Ma after the cessation of Deccan trap volcanism (Fig 5). Very recently, MacIntyre et al [48] have found a peak of magmatic activity ca 63 Ma ago in the Seychelles, which might correspond to the end of Deccan volcanism and/or to the initiation of rifting in the Arabian Sea. This raises the question of a possible causal relationship between the two events If we restore the Indian Ocean to its configuration at the time of the KTB (Fig 7), it is tempting to also restore the Deccan volcanism to a more regular

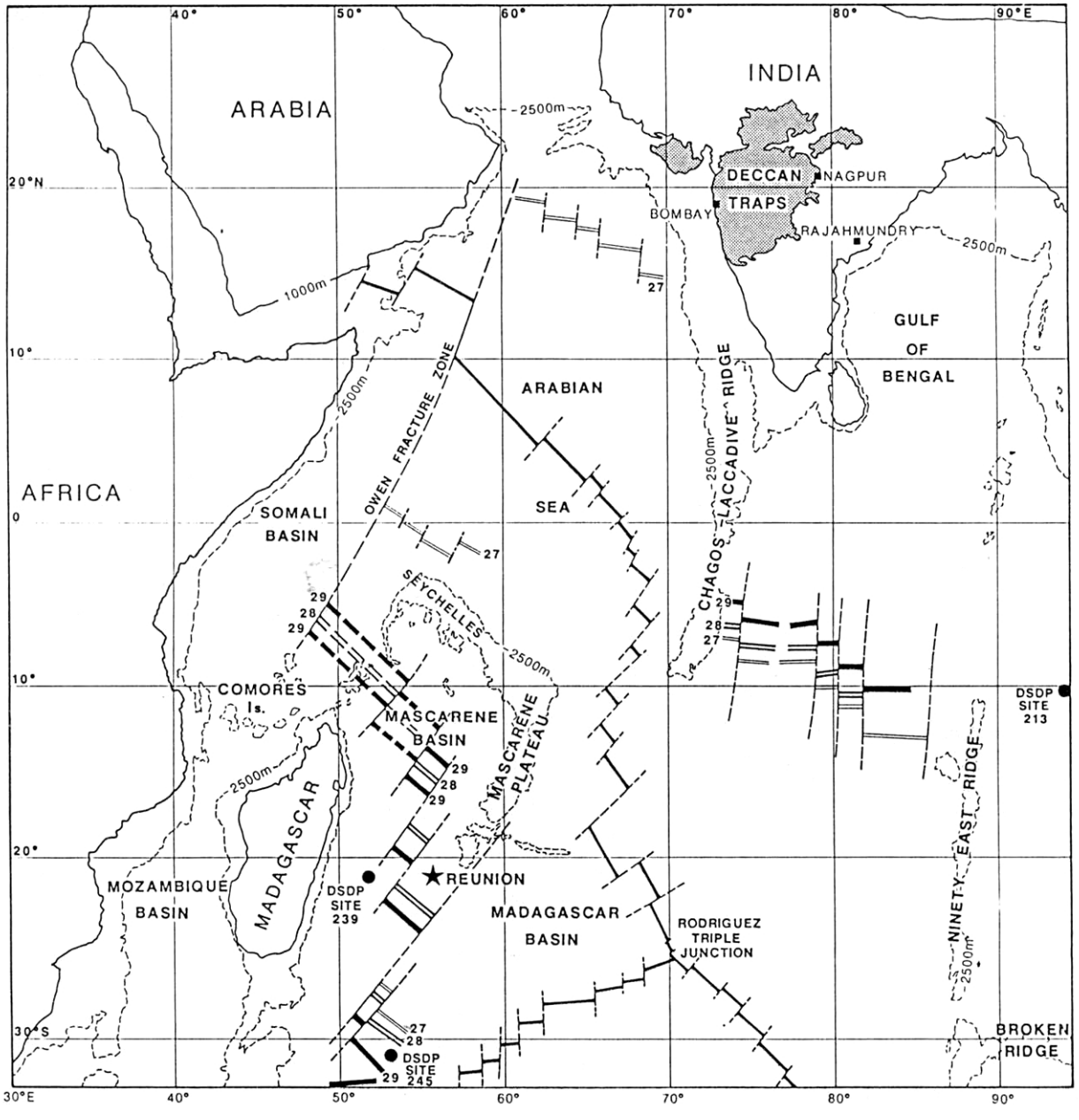


Fig 6 The Deccan traps in the framework of the Indian Ocean Major physiographic features, including mid-ocean ridges and oceanic plateaus, are indicated, together with the trace of magnetic anomalies (chrons) 27, 28 and 29, mostly from Tisseau [47], Patriat [58] and Patriat and Schlich [46] Also shown are DSDP Holes 213, 239 and 245 (numbered asterisks) and the Reunion hotspot (star)

elliptic shape, which would imply that a sizeable fraction of these continental basalts might extend over the bend of the Seychelles-Mascarene structure, although not near the Seychelles Islands proper. Seismic refraction lines [49] show a ~2 km thick layer at depths from 1 5 to 4 km below the Seychelles-Mascarene bend, with velocities of 4.5 km s^{-1} which could correspond to basalt

Alvarez et al [45] suggested that both extinctions and unusually high Ir concentrations might

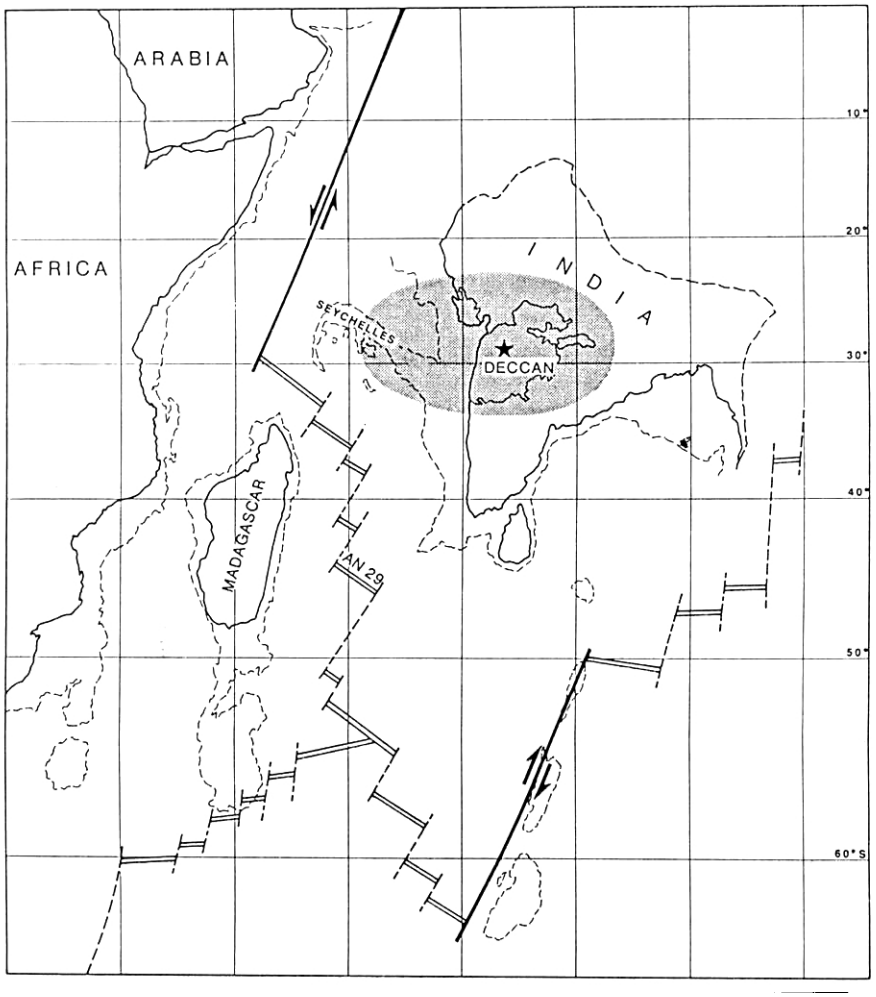


Fig 7 Paleoreconstruction of the Indian continent and surrounding oceanic plateaus and landmasses at the time of chron 29R (i e at the Cretaceous/Tertiary boundary), when the Deccan traps are thought to have erupted The paleolatitudes for India and Madagascar/Africa are from this study and Besse et al [13], and the relative motions are from Patriat [58] The possible original extent of the traps is shaded and the position of the Reunion hotspot is shown as a star (using Africa's absolute motion from Morgan [57])

have been caused by the impact of a ~ 10 km diameter asteroid ("external" hypothesis) Should our correlation of the Deccan traps with the KTB be accepted, it remains to be seen how the volcanism can be related to the impact. One problem with the impact hypothesis is that the locus of impact has yet to be found. Despite tenuous isotopic evidence [50] that would seem to indicate an oceanic site of impact, one might suggest that the Deccan marks this site and represents the equivalent of a lunar mare on Earth Following calculations of O'Keefe and Ahrens [51] we find that an object somewhat larger than generally assumed could on impact have broken through the crust as well as perhaps much of the rigid Indian lithosphere and have extracted over 10^6 km^3 of melt in the subsequent 1 Ma However, tectonic disturbances related to a major asteroid impact have not been described in India around the basalt outcrops It is not clear whether the thermal disturbance in the asthenosphere and lithospheric weakening would have been sufficient to account for the subsequent rifting of the Arabian Sea.

An alternate "internal" hypothesis (presently the view of a minority) has been put forward by Officer and Drake [41,52] and can be traced back

to Vogt [44] It has been noted that the KTB Ir enrichment might not be instantaneous, but rather might span 10^4 to 10^5 years (see also Rocchia et al [53–54]) An "external" explanation involves comet showers, whereas an "internal" one proposes to relate all KTB events to exceptionally intense volcanism Recently, Loper and McCartney [55] have proposed that this (continental flood basalt) volcanism might have been triggered by eruption of hot material from the lower mantle (D″) thermal boundary layer, the exceptional plume that would have triggered the Deccan volcanism could have left the D″ layer at about 80 Ma, following a 30 Ma period of constant magnetic polarity (the Long Normal Cretaceous superchron), thought to be associated with a minimal level of convection in the core. If the plume was of the low-viscosity cavity type described by Olson and Singer [56], it could have provided the thermal and kinematic disturbance required to trigger fusion of the continental lithosphere, fracturing and later rifting of the Arabian Sea in a single event. Besse (in preparation) has interpolated the absolute motion of India in a hot spot reference frame, using Morgan's [57] absolute motion for Africa and recent kinematic models of opening of the Indian Ocean [58], at the time of chron 29 the total angular rotation of Africa in the hot spot reference frame (or "absolute" motion) is 15 3° about a pole located at 43 2°N, 51°E and the total rotation of India with respect to Africa is 45 6° about a pole at 13 1°N, 31.4°E. It is found that the hotspot which is presently located near 21°S under Reunion Island (Fig. 6) stood below the Deccan traps at the time they erupted (Fig. 7) Thus, the Reunion hotspot could be the weak remainder of the plume event that triggered the trap eruptions and subsequently the Chagos, Laccadive and Maldive Islands [59,60].

We do not intend to say that the "external" vs. "internal" KTB controversy is definitely solved, but we wish to point out that many of the KTB anomalies have now received reasonable explanations by proponents of the two views Such is the case for the iridium, which may well come from an asteroid, but might also be internal, as observed by Zoller et al [61] and Olmez et al. [62], for the present-day Kilauea volatiles The high-pressure (shock) metamorphism can indeed by due to an impact, but has also now been found for catastrophic volcanic arc explosions [63], admittedly in a geodynamical setting quite distinct from that of the Deccan Finally, the environmental stresses that are thought to have caused the demise of many Cretaceous species probably involved obscuring of the atmosphere by dust, and acid rain the impact and the volcanic events appear to be equally able to produce both. The case of HNO_3 dominated acid rain linked to an impact has been made for instance by Prinn [64] But one may recall that the 1783 Laki fissure eruption in Iceland [65], which caused the loss of 75% of all livestock and the death of 24% of the Icelanders, was due to a toxic dry haze rich in sulfur and fluorine composites This largely non-explosive eruption produced only 12 km^3 of lava over 8 months, yet injected 10^7 to 10^8 tons of H_2SO_4 in the atmosphere, and resulted in a drop of 5°C in the following north hemisphere winter temperatures [65] It would be interesting to test whether the Deccan lavas were unusually rich in sulfur, and to model the injection of S in the atmosphere related to more than 10^6 km^3 of lava erupted over less than 1 Ma.

Admittedly, much work remains to be done in order to answer some of the questions raised in this study for instance, we need to refine magnetostratigraphic and $^{39}Ar/^{40}Ar$ studies of the Jabalpur and Narmada areas where the oldest flows may be exposed, to acquire more magnetic anomaly data from the Arabian Sea to reconstruct its detailed evolution, to drill the Seychelles-Mascarene bight in order to test for the original extension of the traps westward It is our feeling that our hypothesis that the Deccan eruptions should be linked to the KTB and must be accounted for by KTB scenarii may eventually allow tests of "internal" vs "external" models for this major geological event

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